

Scope

Definition

Scope determines where a variable can be accessed within a program.

A variable is only accessible within the region where it is defined.

Types of Scope

1. Local Scope
2. Global Scope
3. Global Keyword
4. Nonlocal Scope (Nested Functions)

1. Local Scope

A variable declared inside a function is called a local variable.

It can only be accessed inside that function.

Example

Python

```
def student():  
    name = "Teja"  
    print(name)  
student()  
# Output:  
# Teja
```

The variable name exists only inside the function.

2. Global Scope

A variable declared outside all functions is called a global variable.

It can be accessed from anywhere in the program.

Example

```
Python
company = "Codegnan"
def display():
    print(company)
display()
# Output:
# Codegnan
```

Local vs Global Variable

```
Python
name = "Global Name"
def display():
    name = "Local Name"
    print(name)
display()
print(name)

# Output:
# Local Name
# Global Name
```

Local variables get higher priority inside the function.

Global Keyword

Used when we want to modify a global variable inside a function.

Example

```
Python
count = 10
def update():
    global count
    count = 20
update()
print(count)
# Output:
# 20
```

Nonlocal Scope

Used in nested functions. Allows modification of variables from the outer function.

Example

Python

```
def outer():
    count = 10
    def inner():
        nonlocal count
        count += 5
    inner()
    print(count)
outer()
# Output:
# 15
```

LEGB Rule

Python searches variables in the following order:

1. Local
2. Enclosing
3. Global
4. Built-in

Example

Python

```
x = 100
def outer():
    x = 50
    def inner():
        x = 20
        print(x)
    inner()
outer()
# Output:
# 20
```

Python finds x in Local Scope first.

Pass by Value and Pass by Reference

Important Note

Python technically uses **Pass by Object Reference**.

For learning purposes:

- Pass by Value → Immutable Objects
- Pass by Reference → Mutable Objects

Pass by Value

When immutable objects are passed to a function, changes inside the function do not affect the original value.

Immutable Types

- int
- float
- bool
- str
- tuple
- frozenset
- complex

Example

```
Python
def update(number):
    number = 100
    print("Inside Function:", number)
value = 50
update(value)
print("Outside Function:", value)
# Output:
# Inside Function: 100
# Outside Function: 50
```

Original value remains unchanged.

Pass by Reference

When mutable objects are passed to a function, changes inside the function affect the original object.

Mutable Types

- list
- set
- dict

Example

Python

```
def update(items):  
    items.append("Laptop")  
cart = ["Mobile", "Watch"]  
update(cart)  
print(cart)
```

```
# Output:  
# ['Mobile', 'Watch', 'Laptop']
```

The original list changes.

Example with Dictionary

Python

```
def update(profile):  
    profile["city"] = "Hyderabad"  
customer = {  
    "name": "Teja"  
}  
update(customer)  
print(customer)
```

```
# Output:  
# {'name': 'Teja', 'city': 'Hyderabad'}
```

Pass by Value vs Pass by Reference

Feature	Pass by Value	Pass by Reference
Original Data Changes	No	Yes
Immutable Objects	Yes	No
Mutable Objects	No	Yes
Examples	int, str, tuple	list, set, dict

Recursive Functions

Definition

A recursive function is a function that calls itself.

Recursion continues until a stopping condition (Base Case) is reached.

Structure of Recursion

Every recursive function must have:

1. Base Case
2. Recursive Call

Syntax

```
Python
def function():
    if base_condition:
        return
    function()
```

Advantages of Recursion

- Cleaner code
- Suitable for tree structures
- Suitable for graph traversal
- Useful in divide-and-conquer algorithms
- Reduces lengthy loop logic

Disadvantages of Recursion

- More memory usage
- Slower than loops
- Risk of RecursionError if base case is missing

Infinite Recursion

Python

```
def test():  
    test()  
test()
```

This causes: RecursionError: maximum recursion depth exceeded Because there is no base case.

Recursive Function

Q1. Print Numbers from 1 to N

Write a recursive function to print numbers from 1 to N.

Test Case

Input:

5

Output:

1 2 3 4 5

Q2. Print Numbers from N to 1

Write a recursive function to print numbers from N to 1.

Test Case

Input:

5

Output:

5 4 3 2 1

Q3. Sum of N Natural Numbers

Write a recursive function to find the sum of first N natural numbers.

Test Case

Input:

5

Output:

15

Explanation:

$$1 + 2 + 3 + 4 + 5 = 15$$

Q4. Product of N Natural Numbers

Write a recursive function to find the product of first N natural numbers.

Test Case

Input:

5

Output:

120

Explanation:

$$1 \times 2 \times 3 \times 4 \times 5 = 120$$

Q5. Factorial of a Number

Write a recursive function to find the factorial of a number.

Test Case

Input:

5

Output:

120

Explanation:

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

Q6. Power Calculation

Write a recursive function to calculate x raised to the power n.

Test Case

Input:

Base = 2

Power = 4

Output:

16

Explanation:

$$2^4 = 16$$

Q7. Reverse a String

Write a recursive function to reverse a string.

Test Case

Input:

python

Output:

nohtyp

Q8. Count Characters in a String

Write a recursive function to count the number of characters in a string without using len().

Test Case

Input:

Python

Output:

6

Q9. Print String Pattern

Write a recursive function to print the following pattern.

Test Case

Input:

abcdef

Output:

a

ab

abc

abcd

abcde

abcdef

Q10. Print Reverse String Pattern

Write a recursive function to print the following pattern.

Test Case

Input:

abcdef

Output:

abcdef

abcde

abcd

abc

ab

a

Q11. Print Sliding Window Pattern

Write a recursive function to print all substrings of length 4.

Test Case

Input:

abcdef

4

Output:

abcd

bcde

cdef

Q12. Sum of Digits

Write a recursive function to find the sum of digits of a number.

Test Case

Input:

12345

Output:

15

Explanation:

$1 + 2 + 3 + 4 + 5 = 15$

Bonus Q13. Product of Digits

Write a recursive function to find the product of digits of a number.

Test Case

Input:

1234

Output:

24

Explanation:

$1 \times 2 \times 3 \times 4 = 24$

Q14. Count Vowels in a String

Write a recursive function to count vowels in a string.

Test Case

Input:

education

Output:

5

Q15. Check Palindrome

Write a recursive function to check whether a string is a palindrome.

Test Case 1

Input:

madam

Output:

Palindrome

Test Case 2

Input:

python

Output:

Not Palindrome

Q16. Decimal to Binary

Write a recursive function to convert a decimal number to binary.

Test Case

Input:

10

Output:

1010

Q17. Find Largest Digit

Write a recursive function to find the largest digit in a number.

Test Case

Input:

59372

Output:

9

Q18. Fibonacci Series

Write a recursive function to print the nth Fibonacci number.

Test Case

Input:

7

Output:

13

Explanation:

0 1 1 2 3 5 8 13